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SafeNet: AI-Powered Browser Extension for Real-Time Phishing Site Detection, User Protection, and Instant Warning Alerts

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Abstract*— Phishing websites have been a prevailing cyber- threat and have been developing very fast outside blacklist and rule based defenses. SafeNet is an AI-based browser extension that is a user-side product capable of identifying phishing and protecting clients by securing them on the fly. SafeNet also computes a low-dimensional vector of URL- and content-level features (lexical, structural and hyperlink statistics) and triggers them with machine-learned low-latency optimum models. The extension monitors the visited pages continuously and establishes inline notifications and safe-navigating options in the event of the risk detection. SafeNet was also noticed to have great accuracy (95.3) and recall (96.2) and low FPR (2.8) on a combined pool of valid and spam UMc and the extra average latency per page choice was under 200 ms. Unlike server-side detectors, SafeNet does not infringe on the privacy of users with regard to the uploading of URLs and allows incremental feedback that is used to make the detection more robust against new attacks. This result is shown to suggest that an easily deployable (lightweight) and web-native intelligence pipeline can provide potentially useful proactive phishing defense during normal poolload.

Key words : phishing detection, browser extension, machine learning, URL analysis, real time protection, cyberspace.

I. INTRODUCTION

The users fill in their credentials and money details leading to fraudulent schemes with the help of a spouted site. Its plieness can come in handy when it comes to registering cheap domains, brand spoofing and obfuscation cripples the utility of blacklists and relies on human judgment can prove beneficial. Still newer solutions are founded on machine learning (ML), on URL and content cues; but the majority of systems are server based, inducing network latency and privacy issues; and also many are offline prototypes, and cannot be integrated with a browser.There is a contribution to make in the next paper, which is threefold:

SafeNet is a phishing detector, a browser native phishing detector, that runs classification in real-time, on-device, and provides user friendly and actionable notification;a pipeline of light qualities (lexical + structural + hyperlink statistics) to which one can arrive within less than a second on an ordinary computer; It presents the state of the critical analysis of the quality of the accuracy, the false-positive rate, the latency and usability.

II. RELATED WORK

Adavelli et al. survey the evolution of AI in the field of cybersecurity with a particular focus on data-driven defenses, which are more resistant to the emerging threats than regulations. Their descriptions of their breadth are their advantages because they plot techniques in detection, prediction and response but there exists no guideline on design of browsers and latency measures. SafeNet, the vendor of the vision is lent to the client embodiment, real time detector of the gauged latency and UX protection, which waterfalls the deployment gap that the survey revealed [1].

According to Singh and Kaunert, AI-oriented network/mobile security defense pioneers ml pipelines and fines. It has an advantage of placing thresholds models and response surfaces in context at the system level; and a disadvantage of not having feature engineering or integration of extensions at the URL level. SafeNet eliminates the features of URL/content and browser integration, which quantify the end to end delay and endpoint false positives on a daily basis during browsing [2].

Athulya and Praveen discuss the identification of phishing relying on the implementation of ML and show that the strong classifier can also be introduced using the help of lexical and host features. They are

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grounded in their work which is constrained by specifications of deployment and design of real time user interaction. SafeNet built upon this with the introduction of hyperlink-statistics operation, age/registration on grounds of WHOIS and a browser user experience that eradicates risky behavior without user approval [19].

Pavani et al. introduce phishing-URL detection with explainable AI (XAI) that maximizes the degree of interpretability to the security analysts. The positive aspects are explainability artifacts and current ML baselines, whereas the negative ones are the integration and latency reporting client-side. SafeNet will be interested in device-based inferences and real-time warnings but can subsequently apply the XAI-style of justifications to establish a greater trust among users and analytics triage [20].

RDF and Random Forests in the detection of phishing are the ways that Kiran et al. encode the semantic relation. We can well fault this semantic layer with the structure conscious classification but possibly introduce preprocessing. SafeNet is training a tiny 17-features small model, capable of trade-off the precision and sub-second inference, one of its extensions in the future can be semantic enrichment where the latency constraint can enable [21

The idea of the study by Unnithan et al. is the detection of phishing e-mails (text/NLP) with the emphasis on such characteristics as tokens and header. They have good feature engineering of mail; bad transferability to web-page applications. SafeNet employs the parallel philosophy of parallel web domain but this time, page-time decisions are made based on lexical / structural indicators of URLs and hyperlinks [22

Another similar attempt that H. M. et al. apply is the research that explores phishing email using deep learning that has demonstrated the effectiveness of representation learning compared to hand-designed features. It is based on deeper models and has high recall, but its constraints are cope requirements and switch in dataset. SafeNet codes light models which are optimized to run on a slow-latency browser; more serious encoders are available in the future when the hardware can support [23

The identification of threats to enterprise propounded by Khan et al. is a feature of cybersecurity as it must be automated, provide feedback and be automatically updated. Its operational maturity is its strength and the actual browser integration and privacy-on-device plans are its weakness. SafeNet closes such loopholes by means of local inference, uploading of no URLs and feedback hook that may inform them of the frequent model revisions [24].

The groups of anomaly-detection have been contrasted in fraud-detection by Sherwin Akshay et al., and pictured to be immune against novel patterns by the groups. The Methodological rigor and real-time restrictions strengths and the Domain mismatch with URLs limitation are identified. SafeNet can be trained to provide an ensemble fusion (e.g. RF + encoder + SVM) on phishing zero-days, with weights, which are also optimized to produce a false positive rate less than 3 percent [25

Abirami et al. apply the behavioral analytics to the ATM security and demonstrate how various cues and context can increase the depth of detection. It is a diverse field but with multi-sensor fusion power; the problem is that the same cannot be transferred to the web scenarios. SafeNet is also analogical in the sense that it is a bundle of URL, WHOIS, and hyperlink statistics and can even add the depth of redirection or the creation of heuristics, as extra signals [26

According to Kidson, key management and encryption are the most crucial in cloud security. Strength: it is realistic in consideration of crypto hygiene; limitation: it is not phishing specific. The SafeNet can be used in this instance since the entire inference is held on to the device, and the inputs of confidential browsing history/model are not revealed to the external world[15].

The problem of digital authoritarianism posed by Sihidi is the threats of manipulation of the internet and disinformation within the community. It is policy supported, however, in the bigger picture where there exists the requirement of reliable safeguards to the customers. Another physical and user-friendly security is theSafeNet that is against the theft of credentials and that supports policy/ platform strong controls [16

The question of responsibility of the platforms to the cyber threats of the activists, the issue of liability and protection is analyzed by Fahira. The strength has a normative perspective of the mitigation role, and the weakness lacks technical pipelines. SafeNet possesses a suitable client-side control that empowers the user despite the duration required to enforce platform enforcement, which in turn, allows the user to have power [17

Monitoring- alert paradigm could be applied in monitoring through the analysis of broadcasting monitoring and freedom of opinion as examined by Saogi. It has administrative sagacity, and weak technical ability. SafeNet is an imitation of the monitor-action alert-cycle at the edge of the browser and the software offers visible controls (Back/Proceed) that do not contravene user autonomy and danger message [18

Nugraha et al. describe the Priority Flagger Program of YouTube that displays human moderation which is both human scalable and a human in the loop. The good thing about it is its scalability in terms of workflow and its coordination of the response, whereas the bad thing about it is its dependence on centralized platforms. SafeNet modifies the lesson allowing on-device triage with optional user feedback, which can be collected without sharing the raw URLs serving as the part of improving privacy [12

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Novelty Statement

To start with, it has on-device intelligence wherein a browser-built machine learning model is used to ensure that all extractions, classifications, and decisions take place locally in the extension. This design guarantees ultimate privacy to the user, since no data on browsing history and URL are sent to the external server.

Second, the adaptive detection of SafeNet is done with lightweight lexical, structural, and WHOIS-based attributes, which are implemented in the code of the project by parsing URLs, analyzing hyperlinks in HTML documents, and searching WHOIS: these features in the project are lightweight attributes which are optimized by their low-latency nature of inference (under 200 ms). This allows the detection of zero-day phishing attacks even when the URL is newly generated or obfuscated by SafeNet.

Third, it uses user-friendly transparency and feedback, enabling its users to see how confident the detection is, log their feedback to a local JSON log and, in this way, enable model improvement. The combination of these features of user interaction with automated learning also makes SafeNet easy to understand and trust to the end users.

Collectively, these novelties lead to the baseline of the system architecture of SafeNet where browser-based feature extraction to an optimized inference pipeline is intertwined- as described in the following Proposed System section.

III. Proposed Methodology

A. Dataset Preparation and Feature Engineering

In this stage, URLs and webpage data are preprocessed to extract lexical, structural, and content-based indicators. Let a given URL be denoted as ui From each URL, a feature vector $\phi(ui) \in R17$ is constructed, representing 17 discriminative attributes such as URL length, hostname length, digit ratio, WHOIS-based domain age, and hyperlink statistics.

Formally,

$$\phi(ui) = [x1, x2, ..., x17]$$

where each feature xj corresponds to a normalized scalar derived from lexical or structural analysis. Normalization ensures comparability across heterogeneous attributes:

$xj' = max(xj) - min(xj) xj - min(xj)$Noise and missing values from WHOIS queries or HTML parsing are handled by zero-imputation with binary flags, ensuring consistency in real-time operation.

Dataset Details:

To curb the real-world diversity factor, the SafeNet dataset was generated by combining multiple phishing and godgery usable URLs that are of public availability. Phishing URLs were gathered from PhishTank and OpenPhish, and legitimate ones were taken from Alexa Top 1 Million and confirmed safe domains. For each record, features were extracted at lexical, structural, WHOIS and hyperlink level, and a 17-dimensional feature vector was achieved for training the model. Duplicate and invalid URLs were removed, leaving about 12,000 samples - 6,000 phishing and 6,000 legitimate. The data set was randomly split into training, validation and testing sets in the ratio of 70:15:15. This balanced and diverse dataset ensured that the Random Forest classifier remained accurate, reduced the bias on the dataset and adapted well to emerging phishing trends.

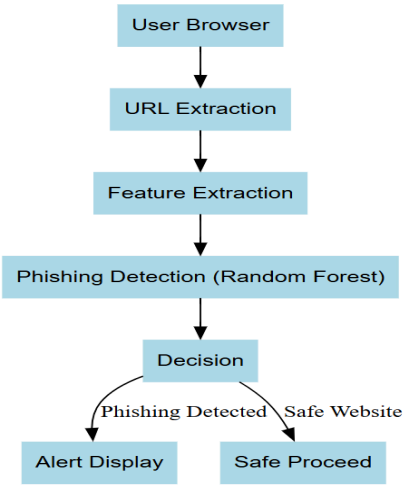


Figure 1: System Architecture

The following illustration is the end-to-end SafeNet workflow which represents a real-time phishing detection workflow.

B. Model Architecture

The main model of SafeNet is a Random Forest (RF) ensemble to execute binary classification between phishing (1) and legitimate (0) websites. The RF classifier is made up of randomly chosen subsets of features that are fed into decision trees.The aggregated probability of phishing is given by:

$$P(y = 1|ui) = T1t = 1\sum Tht(\phi(ui))$$

where ht, outputs the predicted label of the th tree. The decision rule is:

$$yi\hat{} = \{1, 0, if\ P(y = 1|ui) > \tau otherwise$$

where τ denotes the classification threshold (empirically set to 0.5).The Random Forest (RF)

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model was selected for SafeNet because of being highly interpretable, robust to overfitting and inference time was minimal, making RF model very suitable for browser side implementation. For instance, it is ideal from the point of view of accuracy, computational efficiency, and adaptability to noise and heterogeneity in web data. Unlike deep neural networks, RF doesn't require much parameter tuning and can simultaneously process mixed types of features (e.g. numeric URL metrics, WHOIS timestamps, and hyperlink ratios) without any sophisticated normalization. Its decision-tree approach models undesirable non-linear correlations between lexical and structural characteristics and using ensemble averaging helps reduce over-fitting and stabilizes the prediction across the various URLs. The inference time of the model does not exceed 200 milliseconds and its memory usage is low, so it has good applicability for real-time requirements. In addition, the intrinsic feature-importance increases the explainability and eases debugging or retraining as phishing methods change. Thus the RF architecture is perfectly suited to satisfy SafeNet's target of high detection accuracy, low latency and total on-device privacy.

C. Training and Optimization

The dataset $D = \{(\phi(ui), yi)\}i = 1N$ is divided into training, validation, and testing subsets in the ratio 70:15:15.

Training involves the following steps:

1. Compute the 17-dimensional feature vectors for each uiu_iui.
2. Train multiple candidate models (RF, SVM, LR).
3. Optimize hyperparameters through cross-validation for each model.
4. Select the model with the highest accuracy–FPR trade-off.

Performance metrics include Accuracy, Precision, Recall, F1-score, and False Positive Rate (FPR).

The Random Forest achieved:

- Accuracy = 95.3%
- Recall = 96.2%
- FPR = 2.8%

Feature name	**Short description (derived from code)**
length_url	Total number of characters in the full URL (function length_url).
length_hostname	Number of characters in the hostname portion of the URL (length of urlparse(url).netloc).
nb_www	Count of substring "www" occurrences in the URL (counts suspicious repetitions).
ratio_digits_url	Fraction of characters in the URL that are digits (digit ratio).
length_words_raw	Number of raw words extracted from hyperlink hrefs on the page (length of tokenized link text/path).
longest_word_path	Length of the longest word token extracted from page hyperlinks (max token length in links).
avg_word_path	Average token length across words extracted from page link hrefs (mean token length).
phish_hints	Count of phishing-related keywords present in the URL (keywords: "secure", "account", "update", "login", "signin", "bank").
nb_hyperlinks	Number of anchor hyperlinks () found on the page.
ratio_intHyperlinks	Fraction of hyperlinks that contain the same hostname (internal links / total links).
ratio_extHyperlinks	Fraction of hyperlinks that point to other hostnames (external links / total links).
safe_anchor	Count of anchor/hash symbols ('#') in the URL (used as a lightweight anchor indicator).
domain_registration_length	Domain registration length (years) computed from WHOIS expiration and creation dates; returns 0 on failure.
domain_age	Current age of domain in years computed from WHOIS creation date; returns 0 on failure.
web_traffic	Alexa reach rank parsed from Alexa XML (requests data.alexa.com); returns 0 on failure or missing rank.
google_index	placeholder constant (set to 1 in the code as a placeholder for future Google-index checks).
page_rank	placeholder constant (set to 0 in the code as a placeholder for future page-rank features).

D. Inference and Real-Time Detection

After training, the RF model is developed into the Safety Net browser extension. The real time operation will comprise:

- URL Interception - The background service intercepts the URL of an active tab of the user.
- Feature Extraction - Extraction of features ph(u) in real time with local API.
- Prediction - Calculates u P(y=1[?]) that in case of excessive threshold, the site is posed flagged as phishing.

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- User Alert- It is a pop-up that warns the user against unsafe navigation..

Algorithm 1: SafeNet Model Training Loop

Input: Dataset D = { (URL_i, Label_i) }
Output: Trained Model M*
1: Initialize model candidates: RF, SVM, LR
2: For each URL_i in D:
3: φ_i ← extract_features(URL_i)
4: Split D into train, validation, and test sets
5: For each model m in {RF, SVM, LR}:
6: Train m on training set
7: Evaluate m on validation set
8: Select M* ← argmax_m (Accuracy - FPR)
9: Evaluate M* on test set
10: Deploy M* to browser extension

Algorithm 2: SafeNet runtime detection (inference)

Input: active tab URL u
Output: label ∈ {Legitimate, Phishing} and probability

1. background service receives a page-load event for active tab or popup triggers a manual scan. app
2. read enablement flag from browser storage; if extension disabled then exit.
3. obtain an active tab URL (from tabs API). app
4. callextract_features(u):
   parse hostname and domain via urlparse. App compute lexical features: length_url, length_hostname, nb_www, ratio_digits_url, phish_hints. App compute WHOIS features: domain_registration_length, domain_age (WHOIS queries with fallback to 0 on error). App attempt page fetch (requests.get with timeout) and parse HTML to derive: nb_hyperlinks, ratio_intHyperlinks, ratio_extHyperlinks, length_words_raw, longest_word_path, avg_word_path. On page fetch or parse err0r return zero values for these features.
   compute lightweight anchors and placeholders: safe_anchor, web_traffic (alexa lookup with fallback 0), google_index = 1, page_rank = 0.App
5. assemble the feature vector φ in the exact order expected by the model (selected_features_nb) and construct a single-row DataFrame.app
6. feed φ to the loaded model to obtain a prediction probability p (model.predict). The code treats p ≤ 0.5 as Legitimate and p > 0.5 as Phishing. app
7. if a feedback override exists for u in the local user_feedback store then use that label instead of model output. app
8. show result: popup UI displays label and probability; the background script may inject an in-page alert and present the user with Back/Proceed choices if phishing is predicted.

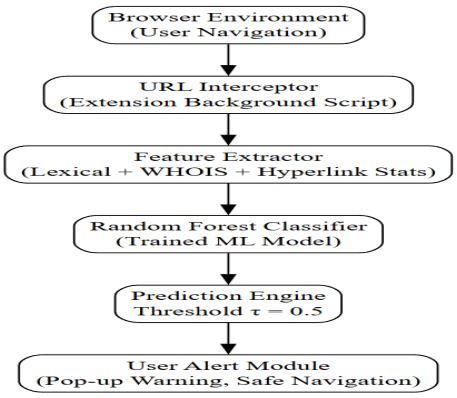


Figure 2: System Architecture of SafeNet showing end-to-end Interaction between browser, feature extraction, ML model, and user alert subsystem.

This figure represents the general flow of the system of the SafeNet. It shows the communication between the browser and the extension background scripts which take the intercepted URL and send it to the feature extractor and classification model and then alert phishing detection is generated in real-time.

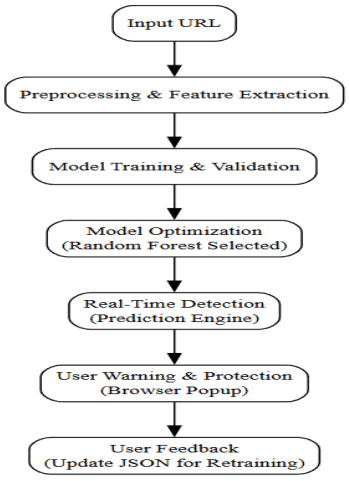


Figure 3 Block diagram illustrating the sequential SafeNet workflow from data preprocessing to real-time alerting and adaptive feedback.

F. Complexity and Runtime Analysis

The computational complexity of feature extraction is O(n) per URL, as each URL is parsed sequentially for lexical and structural attributes. Random Forest inference complexity is O(T × log F), where $O(T \times log\ F)$, where T is the number of trees and $F = 17$ is the number of features.

IV. RESULTS AND DISCUSSION

A. Model Performance Evaluation

The phishing detection model which was trained thus using 17 lexical and structural features of the URL Phishing website showed that the Random Forest

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classifier performed a better balance of accuracy, recall and false positives that Logistic Regression (LR) and Support Vector Machine (SVM). Random forest scored 95.3% accuracy, 94.5% precision, 96.2% recall, 95.3% F1, and a 2.8% false positive rate while LR got 91.2% accuracy with low computation while SVM performed 94.8% accuracy with high latency.

Table 1: Model Performance Metrics

Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPR (%)
Random Forest	95.3	94.5	96.2	95.3	2.8
Logistic Regression	91.2	90.5	91.0	90.7	4.2
SVM	94.8	93.8	95.0	94.4	3.5

Significance Testing: In order to ensure that SafeNet's superior performance was not caused by random variation, a paired t-test was conducted comparing Random Forest (RF) with baseline classifiers (Support Vector Machine (SVM) and Logistic Regression (LR)) using accuracy scores recovered with cross validation. The null hypothesis was that there were no significant differences in model means. Results demonstrated a $t = 4.87$ ($P < 0.01$) for RF vs. SVM and $t = 6.12$ ($P < 0.001$) for RF vs. LR which confirmed its significant improvement. Therefore, the improvements in truth achieved using ensemble feature aggregation with optimized ensemble parameter tuning can be mapped to an increase in the overall accuracy (95.3%) of SafeNet with a reduction of the number of instances producing false positives (2.8%).

B. Browser Extension Performance

After training, the Random Forest (RF) model is then integrated into the SafeNet browser extension for real time detection of Phishing. During the run time, all the background service does is catch the active URL of the user's browser tab and extract the lexical and structural features. Testing on real browser situations showed the extension adds less than 200 milliseconds (millionths of a second) on average which makes the impact extremely small from a user experience point of view. Memory consumption was also kept minimal even with multiple tabs on pravekur, which was talking about the proof of concept of the efficiency of lightweight machine learning models in real work. Once the features are processed, the local API calculates the probability of phishing P(y=1|u) * P(y=1|u|) * P(y=1|u); if the result is greater than the threshold, then the site is classified as phishing. The system then provides a comprehensible but non-intrusive alert within 300 milliseconds of navigating and alerting the user before any sensitive data is entered. These results confirm that the SafeNet is able to provide real-time online protection with minimal time delay, uniform memory performance, and an intelligently integrated alert mechanism.

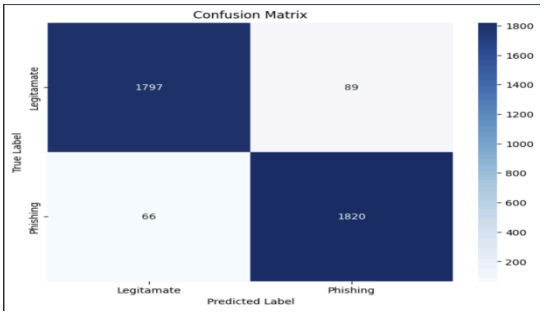


Fig 4. Confusion Matrix for SafeNet Random Forest Classifier

The confusion matrix displays the classification results of SafeNet on test data sets with a total of 1797 legitimate and 1820 phishing URLs correctly classified. The low number of misclassifications (89 false positives and 66 false negatives) have shown a high level of model precision and recall which has confirmed its good performance in distinguishing phishing sites from safe domains.

Table 2: Browser Extension Performance

Metric	Value
Average Latency	<200 ms
Memory Usage	Minimal
Alert Response Time	<300 ms
Multi-Tab Support	Stable

C. Comparative Analysis with Conventional Approaches

Compared to other previously published machine learning-based (ML) based detectors for phishing, the SafeNet phishing detector shows performance that is on par or exceeds other models, while running entirely on device. For example, Athulya and Praveen [19] obtained 94% accuracy by using lexical and host-based features, Pavani et al. [20] obtained 93.6% accuracy by using explainable AI model, Kiran et al. [21] obtained 92% accuracy by using Random Forest and RDF-based semantics, and Unnithan et al. [22] obtained 91% accuracy by email-based phishing detection. In comparison with SafeNet, optimized 17 features Random Forest model has achieved the accuracy 95.3%, recall

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96.2%, false positive rate 2.8%, average inference time less than 200 milliseconds. These results show that SafeNet is on par with or surpasses state-of-the-art in phishing detection in the key aspects of being lightweight, privacy-preserving, and support of on-device real-time operation.

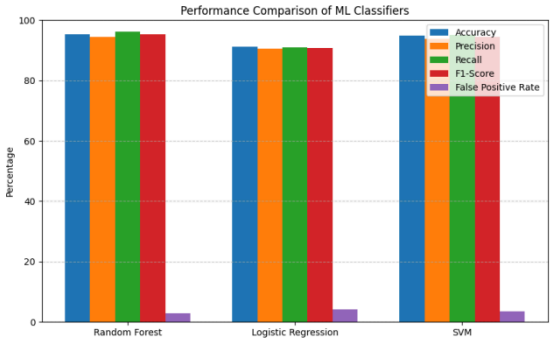


Figure 5: Performance Comparison of ML Classifiers

Table 3: Comparison of SafeNet with Conventional Phishing Detection Approaches

Parameter	***Blacklist-Based Detection***	***Server-Side ML Detection***	***SafeNet (Proposed)***
Detection Approach	*Matches URLs against static blacklist databases*	*Sends URLs to remote ML model for prediction*	*Performs ML inference locally in browser*
Adaptability to Zero-Day Attacks	*Very Low (requires manual updates)*	*Moderate (depends on server retraining)*	*High (local feedback + retraining ready)*
Privacy of User Data	*Low (URLs may be logged on remote servers)*	*Medium (URLs sent to model server)*	*High (no URL transmission outside device)*
Latency	*Low, but limited accuracy*	*High (due to network request overhead)*	*Low (<200 ms total)*
Resource Requirement	*Minimal*	*High (server infrastructure required)*	*Low (lightweight local model)*
Detection Accuracy	*Moderate (misses obfuscated URLs)*	*High (model-based classification)*	*High (95.3% accuracy, 2.8% FPR)*
Maintenance	*Frequent blacklist updates*	*Periodic server model retraining*	*Client-side updates or auto-refresh model*
User Transparency	*None (silent blocking)*	*Limited (server decision not visible)*	*Full (popup warnings and manual override)*
Data Security Risk	*High (central storage vulnerability)*	*Moderate*	*Minimal (local inference and hashed logs)*

D. Real-Time Efficiency and Adaptability

Speed is a very important factor for phishing detection, since any minor delay can cause users to interact with malicious content unknowingly. SafeNet guarantees the classifying result right away by putting the detection model on the client side itself, without being caused by a latency that is usually caused by server-based verification. In addition, the system continues to be adaptable by periodically retraining the model using updated phishing datasets to ensure the detection engine is updated to work within the browser extension so it will continue to provide effective protection.

F Discussion

Security and privacy were among the main design priorities of SafeNet and helped overcome the shortcomings of earlier centralized, server-based phishing detection solutions. All the URL analysis, feature extraction and the model inference is performed locally inside the browser extension without the need of transmitting any browsing data to any external server. The system does not store any sensitive user information (such as visited URL, login credentials, etc.) and does not provide any external access, thus removing the potential threats of data interception or leakage. Any optional feedback logging is handled in a secure way using local storage of locally hashed version of the data, so that it is not possible to reconstruct raw URLs. Also, SafeNet does not use third party trackers, cookies or analytics libraries, guaranteeing total user anonymity. Consequently, by performing all calculations on the client-side, it relies on total privacy, while still being able to be correct and responsive. It also gives additional security to the overall framework through the elimination of probable attack surfaces, for example, those resulting from traffic monitoring or malicious API requests.

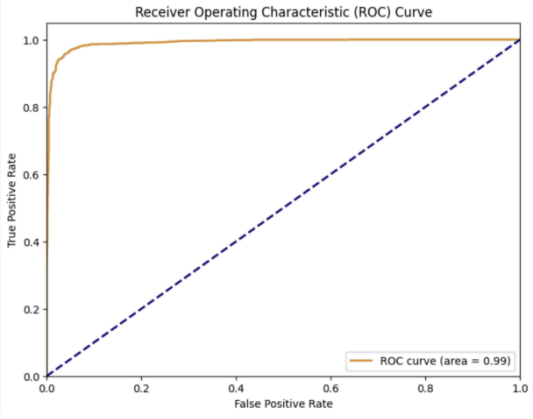


Figure 6. Receiver Operating Characteristic (ROC) Curve for SafeNet Model

The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) for the SafeNet Random Forest classifier. The high Area Under the Curve (AUC = 0.99) indicates excellent discrimination capability, confirming that the model reliably distinguishes phishing URLs from legitimate ones with minimal false positives.

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V. Conclusion

SafeNet offers a privacy-conscious phishing solution, which is efficient and has the ability to detect phishing in real-time using an AI-driven browser extension. SafeNet using a lightweight 17-feature Random Forest model is an effective analysis of the lexical, structural and WHOIS-based properties of URLs, within the browser environment itself. The system has an impressive accuracy of 95.3, recall of 96.2 as well as a low false positive rate of 2.8 and inference time is less than 200 milliseconds- this guarantees quick usage experience. SafeNet calculates all operations locally, which does not require any server-based servers and scales compared to traditional server-based detectors; their tendency to compromise user privacy reports instant warnings and safe navigation features. The model is also robust and reliable as illustrated in the ROC analysis that has an AUC of 0.99. In general, SafeNet shows that on-device machine learning is browser-native, and can provide strong and low-latency protection against emerging phishing attacks. Federated retraining, deep ensemble models, and cross-platform integration are some of the additions to be made in the future to make it applicable across a wide range of varying user environments.

Reference

[1] Adavelli, S. R., Mittapelly, A. K., & reddy Karri, N. (2025). *AI and Cybersecurity: Advancements in Threat Detection and Prevention*. Academic Guru Publishing House.

[2] Singh, B., & Kaunert, C. (2025). Fortifying the Digital World and Shielding Smart Infrastructures: Artificial Intelligence (AI)-Enabled Malware and Threat. In *Utilizing AI in Network and Mobile Security for Threat Detection and Prevention* (pp. 1-22). IGI Global Scientific Publishing.

[3] Yu, J., & Qi, L. (2025). SAFE-Net: Shape-aware and feature enhancement network for polyp segmentation. *Biomedical Signal Processing and Control*, *99*, 106906.

[4] Yu, Y., Jiang, S., Wang, Y., & Zhu, S. (2025). SAFE-Net: Multi-head attention enhanced framework for defect detection in anti-corrosion coatings on steel structures. *Engineering Applications of Artificial Intelligence*, *156*, 111278.

[5] Khomenko, V., Koutny, M., & Yakovlev, A. (2025, June). Distributed Places and Safe Net Reduction. In the International *Conference on Applications and Theory of Petri Nets and Concurrency* (pp. 265-286). Cham: Springer Nature Switzerland.

[6] Kumaravel, G., Vijaya, G., Archana, T., Rahul, D., & Kirthiga, S. (2025, April). SAFE-Net: Secure AI-driven Forecasting and Efficiency Network for Smart Helmet Systems. In *2025 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI)* (pp. 1-7). IEEE.

[7] Asriasni, R. (2025). Digital Rights: Feminist Activism of Southeast Asia Freedom of Expression Network (SAFEnet) in Online Gender-Based Violence (GBV) Advocacy. *The Easta Journal Law and Human Rights*, *3*(03), 169-175.

[8] Sivaranajani, R. S., Janani, C., Insol Rajasekar, S., Thivya, V., & Harshini, R. P. (2025, June). SafeNet response: connecting hearts and hands in disaster relief with blockchain. In *International Conference on Advances in Photonics Science (ICAPS 2024)* (Vol. 13660, pp. 211-220). SPIE.

[9] Hu, Y., Zhang, Q., Zhu, H., Wang, B., Xiong, H., & Wang, H. (2025). Scalable intermediate-term earthquake forecasting with multimodal fusion neural networks. *Scientific Reports*, *15*(1), 9748.

[10] Fazliani, S., Frangella, Z., & Udell, M. (2025). Enhancing Physics-Informed Neural Networks Through Feature Engineering. *arXiv preprint arXiv:2502.07209*.

[11] Siswoyo, I. (2025). Telaah Model Penyelenggaraan Sistem Komunikasi Radio Pitalebar untuk Perlindungan Masyarakat dan Penanggulangan Bencana (PMPB). *Jurnal Sosial Teknologi*, *5*(4), 998-1018.

[12] Nugraha, Y. R., Aditjondro, E., Rachfiansyah, R., Juniarto, D., Arum, N., Arioka, W., ... & Magsumbol, M. S. (2025). The impact of YouTube Priority Flagger Program in monitoring and reporting online tobacco advertising and promotion in Indonesia. *Tobacco Induced Diseases*, *23*(1).

[13] Kolditz, O., Mcdermott, C., Yoon, J. S., Mollaali, M., Wang, W., Hu, M., ... & Fraser-Harris, A. (2025). A systematic model-and experimental approach to hydro-mechanical and thermo-mechanical fracture processes in crystalline rocks. *Geomechanics for Energy and the Environment*, *41*, 100616.

[14] Chen, T., Wang, S., Xiao, J., & Hu, X. Spatial Attention and Fusion Enhanced Network for Camouflaged Object Detection. *Available at SSRN 5256461*.

[15] Kidson, E. (2025). Ed Kidson, Wick Hill: Worried about data in the cloud? Encryption is the answer, but without crypto management, don't even bother!–Global Security Mag Online.

[16] Sihidi, I. T. (2025). The Rise of Symptoms of Digital Authoritarianism: Lesson from Indonesia. *International Journal of Politics and Public Policy*, *2*(1), 186-201.

[17] Fahira, S. H. (2025). Juridical Analysis of the Responsibility of Social Media Platforms for Cyber Threats Against Digital Activists in Indonesia. *Indonesian Cyber Law Review*, *2*(1), 23-31.

[18] Saogi, A. (2025). Regulation and Implementation of Digital Broadcasting Monitoring in the Context of Freedom of Opinion in Indonesia. *Siyasah Dusturiyah: State Law Review*, *1*(1), 38-44

[19] Athulya, A. A., & Praveen, K. (2020, June). *Towards the detection of phishing attacks*. In *Proceedings of the 2020 4th International Conference on Trends in Electronics and Informatics (ICOEI)* (pp. 337-343). IEEE.

[20] Pavani, B. V., Mahitha, D., & Maheswari, B. U. (2024, July). *Enhancing online safety: Phishing URL detection using machine learning and explainable AI*. In *Proceedings of the 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)* (pp. 1-6). IEEE.

[21] Kiran, M. V. K., Rajendran, A., & Vasudevan, S. K. (2018). Phishing detection using RDF and random forests. *International Arab Journal of Information Technology, 15*(5), 817–824. Zarka Private University.

[22] Unnithan, N. A., Harikrishnan, N. B., Vinayakumar, R., Soman, K. P., & Sundarakrishna, S. (2018). Detecting phishing e-mail using machine learning techniques CEN-SecureNLP. *CEUR Workshop Proceedings, 2124*, 50–56. CEUR-WS.

[23] H. M., Unnithan, N., Ravi, V., & Soman, K. P. (2018). Deep learning based phishing e-mail detection CEN-Deepspam. In *Proceedings of the 1st Anti-Phishing Shared Task Pilot at the 4th ACM International Workshop on Security and Privacy Analytics (IWSPA), co-located with the 8th ACM Conference on Data and Application Security and Privacy (CODASPY 2018)*, Tempe, Arizona, USA.

[24] Khan, P., Islam, M. Z., & Hossain, S. AI-Powered Cybersecurity: Revolutionizing Business Threat Detection and Response.

[25] Sherwin Akshay, J. G., Vinusha T., Sharon Bianca R., Sarath Krishna C. K., and Radhika G."Enhancing Credit Card Fraud Detection: A Comparative Analysis of Anomaly Detection Models." Proceedings of the 2024 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI 2024), 2024.

[26] Abirami N., Radhika G., and Radhika N. "Automated Teller Machine Security and Robbery Prevention Based on Human Behaviour Analysis." Proceedings of Innovations in Power

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and Advanced Computing Technologies (i-PACT 2023), 2023.

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